

# NALAN BABURAJAN

Software Engineer — Full-Stack Development, Applied AI & Systems Engineering

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## EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat 2024 – 2028

Bachelor of Technology in Computer Science and Engineering

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Computer Networks, Object-Oriented Programming

Indian School Sohar, Sultanate of Oman 2012 – 2024

Senior Secondary Education (PCM-IP)

CBSE

## EXPERIENCE

AI Intern — VaultStack AI June 2026 – Present

- Building full-stack, on-premise AI systems that let enterprise clients automate workflows within their own secure infrastructure, cutting reliance on third-party cloud AI APIs.
- Working across agentic pipelines (LangGraph, AutoGen) and local LLM inference to containerized deployment (Docker, FastAPI), helping ship production-ready systems for enterprise clients.

## TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, C, C++, SQL

Web & Backend: Next.js, FastAPI, REST APIs, PostgreSQL, Pydantic v2

AI / ML: TensorFlow/Keras, scikit-learn, LangGraph, AutoGen, RAG pipelines, local/quantized LLMs (Qwen2.5), XGBoost, MediaPipe, OpenCV, Tesseract OCR, spaCy

Tools / Engineering: Git, Docker, Streamlit, Jupyter Notebook, GCP

## PROJECTS

TerraShield Jun 2026 – Jul 2026

Next.js, FastAPI, PostgreSQL, Pydantic (Canara Bank E Suraksha 2.0 Hackathon — 2nd Place)

- Architected a full-stack mortgage fraud-detection platform using Next.js and FastAPI, reducing underwriting turnaround from 3–5 days to under 60 seconds, a 99.9%+ reduction.
- Engineered a strongly typed, asynchronous 5-stage pipeline with Pydantic v2 and 10 REST API endpoints, automating OCR, document forensics, land-record validation, financial analysis, and risk scoring.
- Built a privacy-first AI engine using Tesseract, OpenCV, spaCy, and Qwen2.5 3B to analyze 12 loan-document types while eliminating 100% of paid cloud-AI API calls.

IshaaraAI — Real-Time ISL ↔ Speech Assist System Finalist (300+ teams), IIT Gandhinagar

Python, MediaPipe, TensorFlow/Keras, Vosk

GitHub

- Built a real-time bidirectional communication system enabling interaction between deaf and hearing users, extracting 63 hand landmarks per frame and training a 35-class MLP classifier on a 42,700-image dataset (95–98% accuracy).
- Optimized CPU-only inference to ~20 FPS (50ms/frame) and compressed the model footprint ~200x versus CNN pipelines (11MB vs. multi-GB).

AqrSense — ML-Based Rental Price Prediction System

Python, scikit-learn, Streamlit

Demo — GitHub

- Built an ML pipeline predicting UAE rental prices on 73,742 listings, engineering features down from 17 → 9 and applying Ridge Regression with 5-fold cross-validation (log-RMSE 0.480 ± 0.014).
- Deployed a Streamlit application with real-time inference and geospatial visualization.

## ACHIEVEMENTS

- 2nd Position, Suraksha Cyber Hackathon, Canara Bank — placed 2nd among 9 finalist teams shortlisted from 5500+ teams through multiple rounds of rigorous evaluation, for TerraShield, a full-stack mortgage fraud-detection platform.
- Shortlisted, Amazon ML Summer School 2026 — selected among the top 2.3% of applicants nationally (3,000 out of 130,000+).
- Finalist, Codeversity Hackathon, IIT Gandhinagar — placed among the top 5 teams out of 300+.

## LEADERSHIP

- Anchor, TEDx SVNIT — hosted large-scale academic events and moderated speaker sessions.
- Sponsorship Executive, Mindbend Tech Fest — secured corporate partnership leads for Gujarat’s largest techno-managerial fest.
- Sponsorship Executive, E-Cell SVNIT — contributed to sponsorship outreach and partnership acquisition initiatives.